

Graduation Project — Task Documentation

Person 2 | Report Upload + Document Processing

Backend • Frontend • AI

Powered by LLM + RAG + Agentic AI

Overview

You handle the most critical data entry point of the platform: accepting company reports in any format (PDF, Excel, Word, Image) and using AI to extract structured data from them. Your output feeds every other feature — anomaly detection, RAG, and the dashboard.

Backend Tasks

ASP.NET Core | EPPlus | iTextSharp | Semantic Kernel

1. Database Tables to Create

Table	Key Columns
Reports	Id, FileName, FilePath, DepartmentId, UploadedBy, UploadedAt, Status
ExtractedMetrics	Id, ReportId, CompanyId, Month, Year, KPIName, Value, ConfirmedByManager

2. API Endpoints

Task	Details
POST /reports/upload	Accept multipart file. Save to storage. Trigger background processing.
GET /reports	List all reports for current department/company.
GET /reports/{id}/status	Return processing status: Pending / Processing / Done / Failed.
GET /reports/{id}/preview	Return extracted JSON for manager to review before saving.
POST /reports/{id}/confirm	Manager confirms extracted data. Saves to ExtractedMetrics.
DELETE /reports/{id}	Delete a report and its extracted data.

3. Document Reading Logic

Task	Details
PDF	Use iTextSharp or PdfPig to extract raw text from all pages.
Excel	Use EPPlus to read rows and columns. Convert to key-value pairs.
Word	Use OpenXML SDK to extract paragraph text.
Image	Call an OCR API (Azure Computer Vision) to extract text from image.

AI Tasks — Document Normalization

Semantic Kernel | LLM Prompt Engineering

How It Works

After text is extracted from the file, send it to the LLM via Semantic Kernel with a structured prompt. The LLM reads the text and returns a JSON object containing the KPI values it found. This JSON is then shown to the Manager for review.

Prompt Template to Use

```
You are a business data extraction assistant. Given the following report text, extract ONLY the values for these KPIs: {{kpi_list}}. Return ONLY valid JSON. No explanation. No markdown. If a value is not found, set it to null. Format: {"kpi_name": value, ...} Report text: {{report_text}}
```

Example Input / Output

Input text from report:

```
"January production reached 800 units. Defect rate was 3%. Five employees were absent."
```

Output JSON from AI:

```
{"daily_production": 800, "defect_rate": 3, "absent_employees": 5}
```

Frontend Tasks

Task	Details
Upload Page	Drag-and-drop or file picker. Show upload progress bar. Show processing spinner.
Preview Page	Show extracted JSON as editable form. Manager can fix values before confirming.
Reports List	Table of all uploaded reports with status badge: Pending / Done / Failed.